Appl Math Optim 36:109–123 (1997)

Applied Mathematics & Optimization

Abstract Estimates of the Rate of Convergence for Optimal Control Problems

T. Kärkkäinen and T. Räisänen

Department of Mathematics, Laboratory of Scientific Computing,
University of Jyväskylä, P.O. Box 35, FIN-40351 Jyväskylä, Finland

Communicated by I. Lasiecka

Abstract. A method for solving optimal control problems with general elliptic operators is presented and analyzed. Especially, estimates of the rate of convergence for the control problems with the proposed approach are derived independently of the underlying approximation method. Some numerical experiments with the proposed method are included.

Key Words. Optimal control, Abstract error estimates.

AMS Classification. 65K10, 49M99.

1. Introduction

In this paper we consider control and state-constrained optimal control problems related to an abstract elliptic equation:

$$Ay = f + u \quad \text{in } V'. \tag{1.1}$$

Operator A: $V \to V'$ is assumed to be Lipschitz continuous, pseudomonotone, and coercive. We assume that (1.1) describes some physical process, for which we have a desirable state $\bar{y} \in V_{\mathrm{ad}} =$ *the set of admissible states* $\subset V$ given by a control function $\bar{u} \in U_{\mathrm{ad}} =$ *the set of admissible controls* $\subset U$ through $A\bar{y} = f + \bar{u}$ (in fact, we can have a set of states with $\bar{u}$, because the solution of (1.1) is, in general, not unique). Further, we assume that we have an observation $z \in V$ of state $\bar{y}$. In practice, we can either measure or set values for the observation only at some points in the domain, so actually we have a discrete observation of the form $y(x_i)$, $i = 0, \ldots, n$. After interpolating this point data to whole Ω, we get a distributed observation for $\bar{y}$ which contains interpolation and measurement errors, and also some error due to the size of the solution set with $\bar{u}$. This we model by assuming that between $\bar{y}$ and z we have an observation error of the form

$$\|\bar{y} - z\|_V \leq \varepsilon. \tag{1.2}$$

We can define the following continuous optimal control problem in a standard way:

$$\min_{u \in U_{\text{ad}},\ y \in V_{\text{ad}}} \|y - z\|_V^2 \qquad \text{subject to} \quad Ay = f + u. \tag{1.3}$$

Lemma 2.1 in Section 2 shows the existence of an optimal pair (u^*, y^*) for problem (1.3). Moreover, we have an estimate

$$\|\bar{u} - u^*\|_{V'} + \|\bar{y} - y^*\|_V \leq C\varepsilon. \tag{1.4}$$

Notice that to derive such error estimates is already meaningful, when we have the existence of an optimal pair for (1.3), since (1.4) is valid for all minimizing pairs. Hence, this shows that, with respect to V'- and V-norms, every solution of the continuous optimal control problem is as close to the true solution as the observation error allows. The aim of this paper is to describe a discrete method for the optimal control problem, which will have the same property up to the level of discretization error. This is obtained by including an extra term to the least-squares cost functional, which takes into account the underlying equation (1.1).

Usually a regularization term, which is related to the minimal expenses condition and gives coercivity with respect to the control variable ($\beta\|u\|_U^2$, or more generally $(Nu, u)_U \geq \beta\|u\|_U^2$), is added to the cost functional in (1.3). However, in this case the regularization parameter (or the weight coefficient) β plays a crucial role in the corresponding error estimate. For example, in [9] a general error estimate for linear optimal control problems is obtained, where the error between u^* and the discrete control u^*_{dis} is given by $\beta\|u^* - u^*_{\text{dis}}\|_U$. However, if we let $\beta \to 0$, estimates consisting of terms in such a form say nothing. Moreover, the smaller β is, the bigger error is allowed between u^* and u^*_{dis}. Our approach avoids the use of any regularization parameters by taking into account the constraints in the problem explicitly, and has therefore no such problems. However, as estimate (1.4) suggests, this leads to error estimates between the true control and a computed one which are valid in the dual norm. Only with extra assumptions can we present estimates for higher-order norms.

Theory and applications of optimal control problems and related error estimates are presented in many papers and books, for example, see the books [13], [1], [15], and [17], and the papers [9], [14], [12], and [10]. In this work, we have tried to develop methods for optimal control problems which allow us to establish an abstract theory concerning estimates of the rate of convergence not depending on the actual discretization method used in the computations. Our methods will be applicable with various discretization strategies, especially with the finite element and the spectral methods [8], [4], [6], [16]. Moreover, the operator A is allowed to be very general in our work.

This paper is organized as follows. In Section 2 we recall a few definitions and inequalities needed in the analysis and give existence proofs for the continuous and discrete optimal control problems. In Section 3 we give the analysis of the rate of convergence for the proposed discrete method and some examples where it can be applied. Finally, in Section 4 we present results from numerical experiments, which are computed with the proposed methods.

2. Notations and Preliminaries

The standard notations for Sobolev spaces and associated norms are used. We do not include the domain Ω in the spaces and norms, since we assume it to be always fixed. We use $(\cdot,\cdot)$ to denote the L^2-inner product on Ω and $\langle\cdot,\cdot\rangle$ the duality pairing between V' and V. We regard C as a generic constant which is independent of the discretization parameters.

We denote by N and M the discretization parameters in the discrete spaces. In the finite element case, N (resp. M) corresponds to the diameter of each element of the triangulation of the domain Ω (i.e., $h \simeq 1/N$) and in the spectral methods the degree of the global (defined on the whole Ω) algebraic polynomial. Results concerning the approximation properties of both methods can be found in [7], [8], [4], and [6] and in a combined form in [16].

We use the following inequality:

$$\text{Let} \quad a, b \in \mathbb{R}. \quad \text{Then, for} \quad \alpha > 0, \qquad a\,b \le \frac{1}{4\alpha}a^2 + \alpha b^2. \tag{2.1}$$

By using (2.1) with $\alpha = \frac{1}{2}$ it is easy to prove that for $v_1, \ldots, v_m \in X$ we have an estimate

$$\|v_1 + \cdots + v_m\|_X^2 \le m\,(\|v_1\|_X^2 + \cdots + \|v_m\|_X^2). \tag{2.2}$$

The dual space V' is equipped with a natural norm

$$\|v\|_{V'} = \sup_{\substack{\psi\in V\\ \psi\neq 0}} \frac{|\langle v, \psi\rangle|}{\|\psi\|_V}. \tag{2.3}$$

A direct consequence of this definition for $v \in V'$ and $\psi \in V$ is the inequality

$$|\langle v, \psi\rangle| \le \|v\|_{V'}\|\psi\|_V. \tag{2.4}$$

We recall the basic definitions of the operators we are considering (see, e.g., [18]). Operator A: $V \to V'$ is called *pseudomonotone* if

$$\begin{aligned} &y_n \rightharpoonup y,\ \overline{\lim_{n\to\infty}}\,\langle Ay_n, y_n - y\rangle \le 0 \\ &\quad \Rightarrow \quad \langle Ay, y - v\rangle \le \underline{\lim_{n\to\infty}}\,\langle Ay_n, y_n - v\rangle, \qquad \forall v \in V. \end{aligned} \tag{2.5}$$

A is *coercive* if

$$\lim_{\|y\|_V\to\infty} \frac{\langle Ay, y\rangle}{\|y\|_V} = +\infty. \tag{2.6}$$

Finally, A is *Lipschitz continuous* if

$$\|Av_1 - Av_2\|_{V'} \le C\|v_1 - v_2\|_V, \quad \forall v_1, v_2 \in V. \tag{2.7}$$

Notice that the so-called *hemicontinuity*, i.e., continuity of $t \mapsto \langle A(y+tv), w\rangle$ for all $t \in [0,1]$ and $\forall y, v, w \in V$, follows immediately from the Lipschitz continuity.

Next we consider the existence of a solution for the continuous optimal control problem:

$$\min_{u \in U_{\mathrm{ad}},\ y \in V_{\mathrm{ad}}} J(u,y) = \|y - z\|_V^2 \qquad \text{subject to} \tag{2.8}$$

$$Ay = f + u \quad \text{in } V'. \tag{2.9}$$

Let $A\colon V \to V'$ be a pseudomonotone, coercive, and Lipschitz continuous operator. Our theory applies to quasi-linear operators of the form $-\sum_{i=1}^{d} D_i(F_i(\nabla y) D_i y) + g(y)$ in case we have suitable monotonicity conditions for F_i, and coerciveness and growth conditions for both F_i and $g(y)$. For a given $f + u \in V'$ there exists a solution y of (2.9). Only if A is strongly monotone do we have uniqueness as well. Let V be a Hilbert space and let U be a reflexive Banach space with compact embedding $U \subset V'$. Let $\bar{y} \in V_{\mathrm{ad}}$ be a desirable state given by the control $\bar{u} \in U_{\mathrm{ad}}$ through $A\bar{y} = f + \bar{u}$. We assume that the set U_{ad} of admissible controls is closed, convex, and bounded, and the set V_{ad} is of admissible states is closed and convex. Moreover, the state constraints are assumed to have meaning in the following sense:

$$V_{\mathrm{ad}} \subset V_{\mathrm{att}} = \{y \in V \mid \exists u \in U_{\mathrm{ad}} \text{ s.t. } Ay = f + u \text{ in } V'\}. \tag{2.10}$$

Now we are ready to state our first result in:

Lemma 2.1. *There exists a solution* (u^*, y^*) *for problem* (2.8), (2.9). *If A is linear, the solution* (u^*, y^*) *is unique. Between* $(\bar{u}, \bar{y})$ *and* (u^*, y^*) *we have the error estimate*

$$\|\bar{u} - u^*\|_{V'} + \|\bar{y} - y^*\|_V \le C\varepsilon.$$

Proof. Let $\{u_n\} \subset U_{\mathrm{ad}}$ be a minimizing sequence. Because U_{ad} is closed, convex, and bounded (and therefore weakly compact) in U, there exists a subsequence (denoted by the same symbol) and $u \in U_{\mathrm{ad}}$ s.t. $u_n \rightharpoonup u$ in U. Let $\{y_n\}$ be the sequence of corresponding solutions to (2.9), i.e., $Ay_n = f + u_n$. Because $J(u,y)$ is coercive with respect to y and there exists $\bar{y} \in V,\ A\bar{y} = f + \bar{u}$, for which (2.8) is finite ($\|\bar{y} - z\|_V \le \varepsilon$), also $\{y_n\}$ is bounded, and there exists $y \in V_{\mathrm{ad}}$ s.t. $y_n \rightharpoonup y$ in V. Since the embedding $U \subset V'$ is compact, we have $u_n \to u$ and $Ay_n = f + u_n \to f + u$ in V'. Because A is pseudomonotone, it follows that the three conditions

$$y_n \rightharpoonup y, \qquad Ay_n \rightharpoonup f + u, \quad \text{and} \quad \overline{\lim_{n\to\infty}} \langle Ay_n, y_n\rangle \le \langle f + u, y\rangle \tag{2.11}$$

imply $Ay = f + u$ (see Proposition 27.7. in [18]). Because of the strong convergence $Ay_n \to f + u$, the conditions in (2.11) are valid in our case. Hence, y solves (2.9). The optimality of the pair (u, y) follows from the weakly lower semicontinuity of J and therefore we can redenote it by (u^*, y^*). Moreover, if A is linear, problem (2.9) has a unique solution and $J(u,y)$ becomes strictly convex with respect to y. This gives the uniqueness in this case.

We prove the error estimate. From (2.9), (1.2), and Lipschitz continuity of A it follows that

$$\begin{aligned} \|\bar{y} - y^*\|_V &\leq \|\bar{y} - z\|_V + \|z - y^*\|_V \leq 2\varepsilon, \\ \|\bar{u} - u^*\|_{V'} &= \|A\bar{y} - Ay^*\|_{V'} \leq C\varepsilon. \end{aligned} \tag{2.12}$$

This shows the estimate. $\square$

Next we consider a discrete method to solve the optimal control problem corresponding to (2.8) and (2.9). Let $V_{\text{ad}}^N \subset V$ and $U_{\text{ad}}^M \subset U$ be closed, convex (and U_{ad}^M bounded) approximations of admissible sets V_{ad} and U_{ad}. We point out that it is not necessary to have $V_{\text{ad}}^M \subset V_{\text{ad}}$ and $U_{\text{ad}}^M \subset U_{\text{ad}}$. The discrete optimal control problem reads as

$$\min_{u_M \in U_{\text{ad}}^M,\ y_N \in V_{\text{ad}}^N} J(u_M, y_N) = \|y_N - z\|_V^2 + \|Ay_N - f - u_M\|_{V'}^2. \tag{2.13}$$

Notice that we treat u_M and y_N as separate variables in (2.13), and do not assume that we solve any actual state equation. For this discrete approach we have the existence of a solution in:

Lemma 2.2. *There exists a solution* (u_M^*, y_N^*) *for problem* (2.13). *If A is linear the solution is unique.*

Proof. Let $\{u_{M,n}, y_{N,n}\} \subset U_{\text{ad}}^M \times V_{\text{ad}}^N$ be a minimizing sequence. Because U_{ad}^M is closed, convex, and bounded, and V_{ad}^N is closed and convex with J being coercive with respect to y_N, there exists a subsequence (denoted by the same symbol) and $u_M \in U_{\text{ad}}^M,\ y_N \in V_{\text{ad}}^N$ s.t. $\{u_{M,n}, y_{N,n}\} \rightharpoonup \{u_M, y_N\}$ in $U_M \times V_N$. Because of the finite-dimensionality of the spaces we have, in fact, a strong convergence. From the Lipschitz continuity of A it follows that $Ay_{N,n} \to Ay_N$, and this gives the continuity of J, which means that (u_M, y_N) is optimal and can be redenoted by (u_M^*, y_N^*). For a linear operator A it is a straightforward calculation to show that the Hessian corresponding to $J(u_M, y_N)$ is positive definite in $V' \times V$. Therefore, the uniqueness follows. $\square$

3. Abstract Estimates of the Rate of Convergence

Now we develop an abstract theory concerning estimates of the rate of convergence for the optimal control problems under consideration. The basic assumption is the existence of an observation z of $\bar{y}$ such that

$$\|\bar{y} - z\|_V \leq \varepsilon. \tag{3.1}$$

Recall the original equation

$$Ay = f + u \quad \text{in } V'. \tag{3.2}$$

In the following theory, it is enough to assume the Lipschitz continuity

$$\|Av_1 - Av_2\|_{V'} \leq C\|v_1 - v_2\|_V, \qquad \forall v_1, v_2 \in V, \tag{3.3}$$

of the operator A: $V \to V'$ in (3.2). Notice that we do not, in fact, need any other assumptions concerning the operator A in this error analysis. Other assumptions are only necessary for the existence of a solution for (3.2).

We recall the discrete method. Let V_N be a finite-dimensional subspace of V and U_M of U. With these discrete spaces we minimize a cost functional

$$J_1(u_M, y_N) = \|y_N - z\|_V^2 + \|Ay_N - f - u_M\|_{V'}^2. \tag{3.4}$$

The optimal control problem to be solved reads

$$\begin{aligned}&\text{find } u_M \text{ in } U_{\text{ad}}^M \text{ and } y_N \text{ in } V_{\text{ad}}^N \text{ such that}\\ &\quad J_1(u_M, y_N) \le J_1(\tilde{u}_M, \tilde{y}_N), \qquad \forall \tilde{u}_M \in U_{\text{ad}}^M, \quad \forall \tilde{y}_N \in V_{\text{ad}}^N.\end{aligned} \tag{3.5}$$

By Lemma 2.2, we denote a solution of this problem by (u_M^*, y_N^*). Between $(\bar{u}, \bar{y})$ and (u_M^*, y_N^*) the following estimate holds.

Theorem 3.1.

$$\|\bar{u}-u_M^*\|_{V'}+\|\bar{y}-y_N^*\|_V \le C\left(\inf_{v_N\in V_{\text{ad}}^N}\|y-v_N\|_V+\inf_{w_M\in U_{\text{ad}}^M}\|u-w_M\|_{V'}+\varepsilon\right). \tag{3.6}$$

Proof. From (3.2) we get the following operator identity:

$$A\bar{y} - Ay_N^* = f + \bar{u} - Ay_N^* = -Ay_N^* + f + u_M^* + \bar{u} - u_M^* \quad \text{in } V', \tag{3.7}$$

where it follows that

$$\bar{u} - u_M^* = A\bar{y} - Ay_N^* + Ay_N^* - f - u_M^* \quad \text{in } V', \tag{3.8}$$

which by (3.3) yields the inequality

$$\begin{aligned}\|\bar{u} - u_M^*\|_{V'} &\le \|A\bar{y} - Ay_N^*\|_{V'} + \|Ay_N^* - f - u_M^*\|_{V'}\\ &\le C\|\bar{y} - y_N^*\|_V + \|Ay_N^* - f - u_M^*\|_{V'}.\end{aligned} \tag{3.9}$$

Since (u_M^*, y_N^*) is a minimizer of (3.5), we have

$$J_1(u_M^*, y_N^*) \le J_1(w_M, v_N), \qquad \forall (w_M, v_N) \in U_{\text{ad}}^M \times \in V_{\text{ad}}^N.$$

By (3.4), (3.2), and using (2.2), (3.1), and (3.3) this gives

$$\begin{aligned}&\|y_N^* - z\|_V^2 + \|Ay_N^* - f - u_M^*\|_{V'}^2\\ &\quad\le \|v_N - z\|_V^2 + \|Av_N - f - w_M\|_{V'}^2\\ &\quad\le 2(\|v_N - \bar{y}\|_V^2 + \|\bar{y} - z\|_V^2) + \|Av_N - A\bar{y} + \bar{u} - w_M\|_{V'}^2\\ &\quad\le 2(\|v_N - \bar{y}\|_V^2 + \varepsilon^2 + \|Av_N - A\bar{y}\|_{V'}^2 + \|\bar{u} - w_M\|_{V'}^2)\\ &\quad\le C(\|v_N - \bar{y}\|_V^2 + \|\bar{u} - w_M\|_{V'}^2 + \varepsilon^2), \qquad \forall (w_M, v_N) \in U_{\text{ad}}^M \times V_{\text{ad}}^N.\end{aligned} \tag{3.10}$$

Hence, from (3.10) we deduce that estimates

$$\begin{aligned}\|y_N^* - z\|_V &\le C\left(\inf_{v_N\in V_{\text{ad}}^N}\|\bar{y} - v_N\|_V + \inf_{w_M\in U_{\text{ad}}^M}\|\bar{u} - w_M\|_{V'} + \varepsilon\right),\\ \|Ay_N^* - f - u_M^*\|_{V'} &\le C\left(\inf_{v_N\in V_{\text{ad}}^N}\|\bar{y} - v_N\|_V + \inf_{w_M\in U_{\text{ad}}^M}\|\bar{u} - w_M\|_{V'} + \varepsilon\right)\end{aligned} \tag{3.11}$$

are valid. From the first estimate in (3.11) we obtain

$$\begin{aligned}\|y_N^* - \bar{y}\|_V &\le \|y_N^* - z\|_V + \|z - \bar{y}\|_V \\ &\le C\left(\inf_{v_N \in V_{\mathrm{ad}}^N} \|\bar{y} - v_N\|_V + \inf_{w_M \in U_{\mathrm{ad}}^M} \|\bar{u} - w_M\|_{V'} + \varepsilon\right).\end{aligned} \tag{3.12}$$

Thus, the result follows as a combination of (3.9)–(3.12). This ends the proof. □

Remark 3.1. Because operator A is coercive, assumption (3.3) can be replaced with a weaker assumption (see Chapter 5 of [7]) of the form: let A be Lipschitz continuous for bounded arguments, i.e., for any ball $B(0, r) = \{v \in V\colon \|v\|_V \le r\}$ there exists a constant $C(r)$ such that

$$\|Av_1 - Av_2\|_{V'} \le C(r)\|v_1 - v_2\|_V, \qquad \forall v_1, v_2 \in B(0, r). \tag{3.13}$$

Remark 3.2. In fact, our method in (3.5) can be considered as a penalty formulation of the optimal control problem (see [13] and [2]), but with a fixed penalty parameter (equal to unity). Usually in penalty formulations, we include the equation error term multiplied with some penalty parameter to the least-squares cost functional. However, because we have assumed that our observation is obtained in V, the two terms in our cost functional are already balanced (with respect to the order of convergence) without any actual penalty parameter.

Yet nothing in the previous analysis prevents the use of a penalty parameter or a sequence of penalty parameters as multipliers for the second term in the cost functional (3.4). The only restriction in this case is the requirement of the uniform boundedness of penalty parameters in $\mathbb{R}^+$. This can be seen by repeating the calculations in Theorem 3.1 with a penalty formulation.

Remark 3.3. If the original equation (3.2) was of the form

$$Ay = f + Bu \quad \text{in } V', \tag{3.14}$$

where B is a given, Lipschitz continuous operator from U to V', i.e.,

$$\|Bu_1 - Bu_2\|_{V'} \le C\|u_1 - u_2\|_U, \qquad \forall u_1, u_2 \in U, \tag{3.15}$$

we could, by repeating the analysis in Theorem 3.1, prove the estimate

$$\|B\bar{u} - Bu_M^*\|_{V'} \le C\left(\inf_{v_N \in V_{\mathrm{ad}}^N} \|\bar{y} - v_N\|_V + \inf_{w_M \in U_{\mathrm{ad}}^M} \|\bar{u} - w_M\|_U + \varepsilon\right). \tag{3.16}$$

We now consider one example where our technique can be applied. Let the equation be given as

$$\begin{aligned}&-\nabla \cdot (b(x)\,\nabla y(x)) + y(x)^3 = f(x) + u(x) \quad \text{in } H^{-1}(\Omega), \\ &y|_{\partial\Omega} = 0,\end{aligned} \tag{3.17}$$

where $\lambda_0 \le b(x) \le \lambda_1$ with bounded positive constants $\lambda_0,\ \lambda_1 \in \mathbb{R}$. We assume that we have the observation z of $\bar{y}$ in H_0^1. Let the control constraints be of the form

$$U_{\mathrm{ad}} = \{u \in L^2 |\ \alpha_0 \le u \le \alpha_1 \text{ a.e. in } \Omega\} \tag{3.18}$$

for $\alpha_0,\ \alpha_1 \in \mathbb{R}$. Approximation for this set is defined in a natural way, for $U_M \subset L^2$, as

$$U_{\text{ad}}^M = \{u_M \in U_M \,|\, \alpha_0 \le u_M(x_i) \le \alpha_1\} \tag{3.19}$$

for all discretization points x_i used to define the discrete function u_M.

Corollary 3.1. *Let $\bar{y} \in H_0^1 \cap H^{r+1}$ and $\bar{u} \in H^s$ for $r \ge 1$ and $s \ge 0$. Let V_N be a finite element space in H_0^1 consisting of r-order polynomials in each element of the triangulation of the domain, and let U_M be a finite-dimensional subspace of L^2 with* $\max\{s-1, 0\}$*-order basis in every element. Then the following estimate of the rate of convergence holds*:

$$\|\bar{u} - u_M^*\|_{-1} + \|\bar{y} - y_N^*\|_{H_0^1} \le C(N^{-r} + M^{-(s+1)} + \varepsilon).$$

Proof. Standard approximation results for the finite elements [8], [16] give

$$\begin{aligned} &\|\nabla(\bar{y} - I_N y)\|_0 \le C N^{-r} \|\bar{y}\|_{r+1}, \\ &\|\bar{u} - I_M u\|_0 \le C M^{-s} \|\bar{u}\|_s, \end{aligned} \tag{3.20}$$

where $I_N y$ and $I_M u$ are interpolants of $\bar{y}$ and $\bar{u}$ in V_N and U_M, respectively. As in Lemma 3 of [9] let $P_M u$ be the L^2-projection of $\bar{u}$ into U_M, which by the definition means

$$(\bar{u} - P_M u, \varphi_M) = 0, \qquad \forall \varphi_M \in U_M. \tag{3.21}$$

Using the definition it is a direct calculation to show that this projection also satisfies the second estimate in (3.20). Moreover, because $U_M \subset L^2$,

$$\forall \psi \in H_0^1 \subset H^1, \quad \exists \psi_M \in U_M \quad \text{s.t.} \quad \|\psi - \psi_M\|_0 \le C M^{-1} \|\psi\|_1. \tag{3.22}$$

By (3.21) this means that, $\forall \psi \in H_0^1$,

$$\begin{aligned} (\bar{u} - P_M u, \psi) &= (\bar{u} - P_M u, \psi - \psi_M) \le \|\bar{u} - P_M u\|_0 \|\psi - \psi_M\|_0 \\ &\le C M^{-1} \|\bar{u} - P_M u\|_0 \|\psi\|_1. \end{aligned} \tag{3.23}$$

Hence, by the definition of the H^{-1} norm we get

$$\begin{aligned} \|\bar{u} - P_M u\|_{-1} &= \sup_{\psi \in H_0^1} \frac{|(\bar{u} - P_M u, \psi)|}{\|\psi\|_1} \\ &\le C M^{-1} \|\bar{u} - P_M u\|_0 \le C M^{-(s+1)} \|\bar{u}\|_s, \end{aligned} \tag{3.24}$$

which proves the result. □

Corollary 3.2. *The same estimate as in Corollary* 3.1 *holds, with the same proof, for the Legendre interpolation* (*see Chapters* 4.4–4.5 *of* [16]). *Moreover, just by changing all assumptions and notations to cover the case of the weighted spaces $H_{0,\omega}^1$, $H_\omega^{-1} = (H_{0,\omega}^1)'$ with the Chebyshev weight function $\omega(x) = \prod_{i=1}^d (1/\sqrt{1 - x_i^2})$, the same estimate is also true for the Chebyshev interpolation* (*see Chapters* 4.1–4.3 *of* [16]).

Remark 3.4. In our example we assumed the existence of a distributed observation z in H_0^1. In many cases, the observation is assumed to be available (and the least-squares cost functional constructed) only in L^2. However, because z is an observation of $\bar{y}$, it is reasonable to assume that if $\bar{y} \in H^{r+1}$, then $z \in H^{r+1}$ also. If this is not true, we can still obtain an observation in H_0^1 after regularization of z with some small extra error.

So, we assume that both $\bar{y}, z \in H^{r+1}$. If I_N is the interpolation operator in V_N, where V_N is a suitable finite element space, we can prove, by using the well-known inverse properties of the finite element method, that, if $\|\bar{y} - z\|_0 \leq \tilde{\varepsilon}$,

$$\begin{aligned}\|\nabla(\bar{y} - z)\|_0 &\leq \|\nabla(\bar{y} - I_N\bar{y})\|_0 + \|\nabla(I_N\bar{y} - I_N z)\|_0 + \|\nabla(I_N z - z)\|_0 \\ &\leq CN^{-r}(\|\bar{y}\|_{r+1} + \|z\|_{r+1}) + CN\|I_N\bar{y} - I_N z\|_0 \\ &\leq CN^{-r}(\|\bar{y}\|_{r+1} + \|z\|_{r+1}) \\ &\quad + CN(\|I_N\bar{y} - \bar{y}\|_0 + \|\bar{y} - z\|_0 + \|z - I_N z\|_0) \\ &\leq CN^{-r}(\|\bar{y}\|_{r+1} + \|z\|_{r+1}) + CN\tilde{\varepsilon}. \end{aligned} \tag{3.25}$$

Thus, in this case the last term ε in the estimate of Corollary 3.1 is replaced with $N\tilde{\varepsilon}$.

Remark 3.5. For example, with the finite element method, another way of dealing with the L^2-observation would be to define a cost functional as

$$\tilde{J}_1(u_M, y_N) = \|y_N - z\|_0^2 + N^{-2}\| - \nabla\cdot(b\,\nabla y_N) + y_N^3 - f - u_M\|_{-1}^2, \tag{3.26}$$

where N^{-2} is for balancing the different amount of differentiation in the two terms. For this cost functional, the error estimate corresponding to Corollary 3.1 is of the form

$$\|\bar{u} - u_M^*\|_{-1} + \|\bar{y} - y_N^*\|_{H_0^1} \leq C(N^{-r} + NM^{-(s+1)} + N\varepsilon). \tag{3.27}$$

Remark 3.6. The same kind of formulation for the optimal control problem can be used if we want to obtain higher-order estimates between $\bar{u}$ and u_M^*. Assume that in the finite element case $V_N \subset H^2$ and, for the sake of simplicity, take $M = N$. Keeping in mind Remark 3.5, we define a cost functional as

$$\bar{J}_1(u_N, y_N) = \|\nabla(y_N - z)\|_0^2 + N^{-2}\| - \nabla\cdot(b\,\nabla y_N) + y_N^3 - f - u_N\|_0^2. \tag{3.28}$$

By assuming that $b \in W^{1,\infty}$, which gives

$$\|\nabla\cdot(b\,\nabla v)\|_0 \leq C\|v\|_2, \qquad \forall v \in H^2, \tag{3.29}$$

a minimization of $\bar{J}_1(u_N, y_N)$ for $\bar{y} \in H^{r+1}$ and $\bar{u} \in H^{r-1}$ yields an error estimate

$$\|\bar{u} - u_N^*\|_0 + \|\bar{y} - y_N^*\|_{H^2} \leq C(N^{1-r} + N\varepsilon). \tag{3.30}$$

This can be proved using calculations similar to those used in Theorem 3.1, because now (3.8) is valid in L^2 due to the regularity of the functions there. Moreover, if U_N is a subspace in H^k for some $k \geq 1$, we can obtain higher-order estimates between $\bar{u}$ and u_M^* simply by using the standard inverse estimates for the finite element method in a similar way as used in (3.25).

Remark 3.7. Remarks 3.4–3.6 are all formulated with the finite element method. The reason for this is the simple fact that in all these remarks we have used the inverse estimate in the analysis. In the finite element method, we lose only one order of convergence when using the inverse estimate, but in the global approximation methods (spectral methods) the loss of order is two (see [6] and [16]). So, these global approximation methods would give one order worse estimates in Remarks 3.4–3.6.

Finally, in some cases we can prove an abstract estimate in the L^2-norm between $\bar{u}$ and u_M^*. We assume that for this result the finite element approximation is used for constructing the discrete spaces (this is because we must again use the inverse inequality).

Theorem 3.2. *Suppose that the result of Theorem* 3.1 *is valid. Moreover, assume that* $U \subset V \subset H^k$ *for some* $k > 0$. *Then the error estimate*

$$\|\bar{u} - u_M^*\|_0 \leq C\Bigg(\inf_{w_M \in U_{\mathrm{ad}}^M} (M^k\|\bar{u} - w_M\|_{U'} + \|\bar{u} - w_M\|_0 + M^{-k}\|\bar{u} - w_M\|_U) \\ + M^k \inf_{v_N \in V_{\mathrm{ad}}^N} \|\bar{y} - v_N\|_V + M^k\varepsilon\Bigg)$$

between $\bar{u}$ *and* u_M^* *holds.*

Proof. Because $U \subset V$, we have $V' \subset U'$ and the result of Theorem 3.1 can be written as

$$\begin{aligned} \|\bar{u} - u_M^*\|_{U'} &\leq C\|\bar{u} - u_M^*\|_{V'} \\ &\leq C\Bigg(\inf_{v_N \in V_{\mathrm{ad}}^N} \|\bar{y} - v_N\|_V + \inf_{w_M \in U_{\mathrm{ad}}^M} \|\bar{u} - w_M\|_{U'} + \varepsilon\Bigg). \end{aligned} \tag{3.31}$$

Because $U \subset L^2$, we can use $(\cdot, \cdot)$ to define the duality pairing $\langle \cdot, \cdot \rangle_{U' \times U}$. Hence, using (2.1) and the inverse inequality for the finite element method gives

$$\begin{aligned} &\|\bar{u} - u_M^*\|_0 \\ &\quad\leq \|\bar{u} - u_M^*\|_{U'}^{1/2} \|\bar{u} - u_M^*\|_U^{1/2} \\ &\quad\leq \frac{1}{4\alpha}\|\bar{u} - u_M^*\|_{U'} + \alpha\|\bar{u} - u_M^*\|_U \\ &\quad\leq \frac{1}{4\alpha}\|\bar{u} - u_M^*\|_{U'} + \alpha(\|\bar{u} - w_M\|_U + C_0\, M^k\|w_M - u_M^*\|_0) \\ &\quad\leq \frac{1}{4\alpha}\|\bar{u} - u_M^*\|_{U'} + \alpha(\|\bar{u} - w_M\|_U + C_0 M^k(\|w_M - \bar{u}\|_0 + \|\bar{u} - u_M^*\|_0)) \\ &\quad\leq C(M^k\|\bar{u} - u_M^*\|_{U'} + \|\bar{u} - w_M\|_0 + M^{-k}\|\bar{u} - w_M\|_U) + \tfrac{1}{2}\|\bar{u} - u_M^*\|_0, \end{aligned} \tag{3.32}$$

where C_0 is the constant in the inverse inequality and α is chosen as $\alpha = (1/2C_0)M^{-k}$, which is allowed for $1 \leq M < \infty,\ k \leq k_0 < \infty$. Since (3.32) is valid for all $w_M \in U_{\mathrm{ad}}^M$, the result follows as a combination of (3.31) and (3.32). □

Remark 3.8. From the estimate in Theorem 3.2 we can derive higher-order estimates between $\bar{u}$ and u_M^* by again applying the inverse inequality.

Remark 3.9. Since w_M should approximate $\bar{u}$ well with respect to different norms in Theorem 3.2, a natural candidate for this purpose is a suitable projection (see [3]). The condition $U \subset V$ seems at first quite a strange relation between the two spaces, but it has in some situations a very simple physical explanation. For example, there is no sense in applying a vertical force on the boundary to an elastic membrane whose vertical displacement is fixed to a given y_0 on $\partial\Omega$ [8]. Mathematically, if $y \in H_0^1$, then naturally the control u should be zero on the boundary as well, which happens if $u \in H_0^1$ also. In the discrete level this is even more obvious, when the trapezoidal rule is applied to the integration. In this case all the boundary nodes will be eliminated from the linear system and this also destroys any effect of the control there.

Remark 3.10. What we need in order to realize our techniques is a way to calculate the dual norm in V'. This can be done for any Hilbert space V, because there always exists an isomorphic identity operator $I = id\colon V \to V'$ such that $\|I\varphi\|_{V'} = \|\varphi\|_V, \forall\varphi \in V$. Notice that, for example, in the case of duality between H_0^1 and H^{-1}, we can take the Laplace operator $-\Delta\colon H_0^1 \to H^{-1}$ as I.

4. Numerical Realization and Computed Examples

We describe the numerical realization of the method introduced in the previous section. To do this, we consider the following minimization problem as an example:

$$\min J(u, y) = \|\nabla(y - z)\|_0^2 + \| - \Delta y + y^3 - f - u\|_{-1}^2 \tag{4.1}$$

over

$$u \in U_{\text{ad}} = \{u \in L^2 |\ \alpha_0 \le u \le \alpha_1 \text{ a.e. in } \Omega\}, \tag{4.2}$$

$$y \in V_{\text{ad}} = \{y \in H_0^1 |\ \alpha_2 \le y \le \alpha_3 \text{ a.e. in } \Omega\}, \tag{4.3}$$

for $\alpha_0, \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$.

The aim is first to discretize this problem using the finite element method (FEM). After that, we describe our algorithm in detail and give the computed results from numerical experiments. By the definition of the dual norm, (4.1) is equivalent to the minimization problem

$$\min_{(u,y) \in U_{\text{ad}} \times V_{\text{ad}}} J(u, y) = \|\nabla(y - z)\|_0^2 + \|\nabla\varphi\|_0^2 \tag{4.4}$$

subject to

$$\begin{aligned} &-\Delta\varphi = -\Delta y + y^3 - f - u \quad \text{in } \Omega,\\ &\varphi = y = 0 \quad \text{on } \partial\Omega. \end{aligned} \tag{4.5}$$

For simplicity, let $\Omega \subset R^2$ be the unit square $[0, 1] \times [0, 1]$. Let T_h be a regular triangulation of Ω where h is the grid parameter corresponding to the length of the triangle edges, and let the x_i, $i = 1, \dots, n$, be the nodal points of T_h. We choose an element base which is used to approximate all functions to be

$$V_h = \{\varphi \in H_0^1(\Omega) \cap C(\bar{\Omega}) \mid \varphi_{|T} \in P_1, \quad \forall\, T \in T_h\} \tag{4.6}$$

with the basis functions v^j, $j = 1, \dots, n$, satisfying $v^j(x_i) = \delta_{ij}$ (Kronecker's delta symbol). In the numerical integration we use the following Lobatto formula:

$$\int_T g\, dx = \frac{\text{meas}(T)}{3}(g(z_1) + g(z_2) + g(z_3)), \tag{4.7}$$

where z_i, $i = 1, 2, 3$, denote the vertices of the corresponding triangle T. Moreover, from now on we let $y, u, f \in R^n$ be vectors containing the nodal values of the corresponding state, control, and right-hand side variables. We define the stiffness matrix

$$K = \left(\int_\Omega \nabla v^i \nabla v^j\, dx\right)_{i,j=1}^n \tag{4.8}$$

and the mass matrix

$$M = \left(\int_\Omega v^i v^j\, dx\right)_{i,j=1}^n . \tag{4.9}$$

Thus, our finite-dimensional problem for (4.4), (4.5) is of the form

$$\min_{(u,y)\,\in\, U_{\text{ad}}^h \times V_{\text{ad}}^h} J_h(u, y) = (y - z)^T K (y - z) + \varphi^T K \varphi \tag{4.10}$$

subject to

$$K\varphi = Ky + My^3 - Mf - Mu, \tag{4.11}$$

for

$$U_{\text{ad}}^h = \{\alpha_0 \le u_i \le \alpha_1,\ i = 1, \dots, n\}, \tag{4.12}$$

$$V_{\text{ad}}^h = \{\alpha_2 \le y_i \le \alpha_3,\ i = 1, \dots, n\}. \tag{4.13}$$

We now introduce our algorithm to this specific problem, but it can be easily modified to handle the more complicated problems introduced in previous sections. The main idea is to split the problem into two subproblems (see [1]).

The Sequential Splitting Algorithm.

1. Initialize $k = 0$, $y_0 = z$.
2. For given y_k,

$$\min_{u_k \in U_{\text{ad}}^h} J(u_k) = \varphi^T K \varphi, \quad \text{subject to}$$

$$K\varphi = Ky_k + My_k^3 - Mf - Mu_k.$$

3. For given u_k,

$$\min_{y_{k+1}\in V^h_{\mathrm{ad}}} J(y_{k+1}) = (y_{k+1}-z)^T K(y_{k+1}-z) + \varphi^T K\varphi,$$

subject to $\quad K\varphi = Ky_{k+1} + My^3_{k+1} - Mf - Mu_k.$

4. Test the convergence. Stop or set $k = k+1$ and go to 2.

In the actual computations, each subproblem in the algorithm was solved by a standard descent method. As an optimization routine we used the algorithm described in [5], which is a conjugate gradient type limited memory quasi-Newton algorithm, intended for large bound constrained optimization problems. In the calculation of the dual norm, an FFT-based solver was applied.

Throughout these computations, y was chosen to be $y = \sin(\pi x_1)\sin(\pi x_2)$. Then, with $f = 0$, the optimal control reads as $u = 2\pi^2 y + y^3$. Following these choices we took $\alpha_0 = 0,\ \alpha_1 = 21,\ \alpha_2 = 0,\ \alpha_4 = 1$ Table 1 contains the L^2-errors $\|u - u_h\|_0$ between the true control and the computed one divided by h^2 with different values of h.

In Table 1 we notice the $O(h^2)$-convergence which is better than the estimated one in Theorem 3.2. This follows from two facts: first, there is superconvergence in the discretization points for φ (see [11]) and with the integration formula (4.7) we take full advantage of this phenomena. In fact, we checked this by computing the L^2 error for $\nabla\varphi$ using the Lobatto formula, and the superconvergence was clearly shown as the convergence was much better than $O(h)$ for the piecewise linear approximation. Hence, because of this, we obtained the L^2-error in Table 1 between u and u_h which is the best possible with the discretization used.

Next we add to y an observation error of the form

$$\varepsilon(x_1, x_2) = 10^{-3}\sin(4\pi x_1)\sin(4\pi x_2), \tag{4.14}$$

which means that initially we have an observation $z = y + \varepsilon$. With this term the error obtained in the control is settled to $\|\varepsilon\|_0$ as expected after the first computation, and this is illustrated in Table 2 and Figure 1.

As we have seen, in the discrete level our approach gives us some interesting possibilities for further development. Separation of the state and control variables simplifies computer implementation and allows us to use more efficient algorithms. A standard method for handling the state constraints is to add a penalization term to the cost functional or it can be solved as a nonlinear constraint in the augmented Lagrangian algorithm (see [15]). In our case the box constraints can be introduced explicitly to the optimization routine.

Table 1

$1/h$	$\|u-u_h\|_0/h^2$
8	8.076
16	8.113
24	8.113
32	8.115

Table 2

$1/h$	$\|u-u_h\|_0$
8	0.180
16	0.153
24	0.155
32	0.156

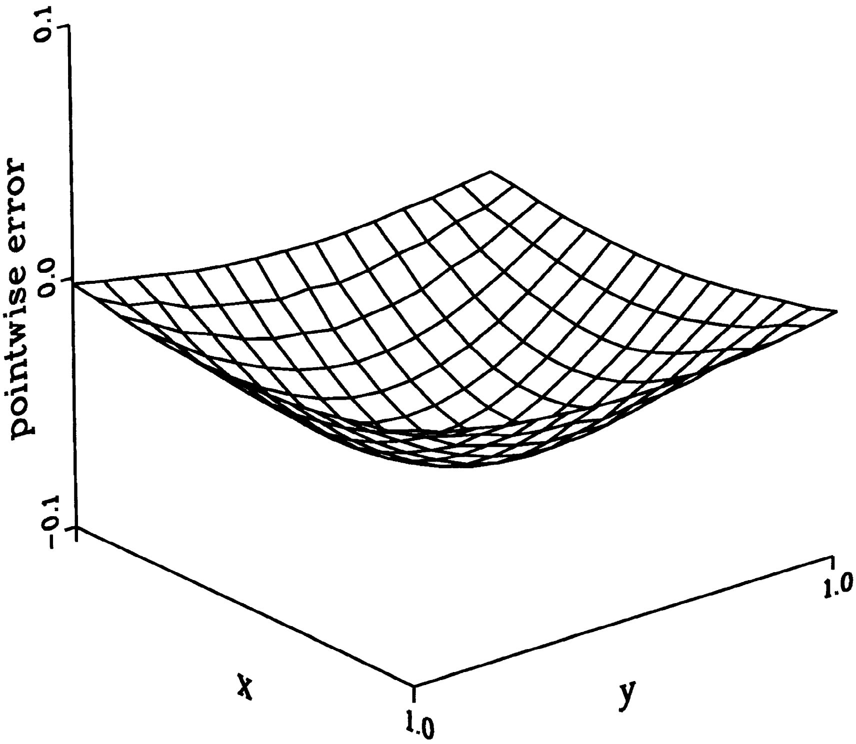

Figure 1. Pointwise error in Table 2 for $h^{-1} = 16$.

For a Lipschitz continuous operator, the standard approach may lead to a nonsmooth optimization problem (see, e.g., [15]) which is avoided here. The use of the equation error term as part of the cost functional allows us to linearize the whole optimization process, even if our model problem is nonlinear. Also, calculation of the gradient of the cost functional in the sequential variable splitting algorithm is simpler. The gradient can be computed directly, without solving any adjoint state equations. These things mean a considerable saving of CPU-time in large-scale optimization problems.

5. Conclusions

A new method for solving control and state constraint optimal control problems together with error estimates was presented. Numerical experiments gave promising results and encourages us to continue this subject. Further studies of the approach developed includes the extension of the idea to the parabolic optimal control problems as well. An important thing is to compare the presented algorithm with classical approaches in order to see how competitive in terms of CPU-time our algorithm is in practical situations.

Acknowledgment

We would like to thank the unknown referee for his valuable comments, which improved the contents of this paper significantly.

References

1. H. T. Banks and K. Kunisch (1989) Estimation Techniques for Distributed Parameter Systems, volume 1 of Systems & Control: Foundations & Applications. Birkhäuser, Boston
2. D. P. Bertsekas (1982) Constrained Optimization and Lagrange Multiplier Methods. Computer Science and Applied Mathematics. Academic Press, New York
3. J. H. Bramble and R. Scott (1978) Simultaneous approximation in scales of Banach spaces. Math. Comp., 32:947–954
4. S. C. Brenner and L. R. Scott (1994) The Mathematical Theory of Finite Element Methods, volume 15 of Texts in Applied Mathematics. Springer-Verlag, New York
5. R. H. Byrd, P. Lu, J. Nocedal, and C. Zhu (1994) A limited memory algorithm for bound constrained optimization. Technical Report NAM-08, Department of Electrical Engineering and Computer Science, Northwestern University
6. C. Canuto, M. Y. Hussaini, A. Quarteroni, and T. A. Zang (1988) Spectral Methods in Fluid Dynamics. Springer Series in Computational Physics. Springer-Verlag, New York
7. P. G. Ciarlet (1978) The Finite Element Method for Elliptic Problems. North-Holland, Amsterdam
8. P. G. Ciarlet and J. L. Lions, editors (1991) Finite Element Methods (Part 1), volume 2. Elsevier, North-Holland, Amsterdam
9. R. S. Falk (1973) Approximation of a class of optimal control problems with order of convergence estimates. J. Math. Anal. Appl., 44:28–47
10. D. A. French and J. T. King (1991) Approximation of an elliptic control problem by the finite element method. Numer. Funct. Anal. Optim., 12:299–314
11. M. Křížek and P. Neittaanmäki (1990) Finite Element Approximation of Variational Problems and Applications, volume 50 of Pitman Monographs and Surveys in Pure and Applied Mathematics. Longman, Essex
12. I. Lasiecka (1984) Ritz–Galerkin approximation of the time optimal boundary control problem for parabolic systems with Dirichlet boundary conditions. SIAM J. Control Optim., 22:477–500
13. J. L. Lions (1971) Optimal Control of Systems Governed by Partial Differential Equations, volume 170 of Grundlehren der mathematischen Wissenschaften in Einzeldarstellungen. Springer-Verlag, Berlin
14. K. Malanowski (1981) Convergence of approximations vs. regularity of solutions for convex, control-constrained optimal-control problems. Appl. Math. Optim., 8:69–95
15. P. Neittaanmäki and D. Tiba (1994) Optimal Control of Nonlinear Parabolic Systems; Theory, Algorithms, and Applications. Pure and Applied Mathematics. Marcel Dekker, New York
16. A. Quarteroni and A. Valli (1994) Numerical Approximation of Partial Differential Equations. Number 23 in Springer Series in Computational Mathematics. Springer-Verlag, Berlin
17. D. Tiba (1995) Lectures on the Optimal Control of Elliptic Equations. Lecture Notes 32. Department of Mathematics, University of Jyväskylä
18. E. Zeidler (1990) Nonlinear Functional Analysis and its Applications; II/B: Nonlinear Monotone Operators. Springer-Verlag, Berlin

Accepted 14 *February* 1996